MSG No. 190.228 $P\bar{6}2c1'$ [Type II, hexagonal]

Table 1: Wyckoff site: 2a, site symmetry: 32.1'

No.	position	mapping
1	$[0, 0, 0]$	[1,2,3,4,5,6,13,14,15,16,17,18]
2	$\left[0, 0, \frac{1}{2}\right]$	[7,8,9,10,11,12,19,20,21,22,23,24]

Table 2: Wyckoff site: 2b, site symmetry: -6..1'

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	[1,2,3,7,8,9,13,14,15,19,20,21]
2	$\left[0, 0, \frac{3}{4}\right]$	[4,5,6,10,11,12,16,17,18,22,23,24]

Table 3: Wyckoff site: 2c, site symmetry: -6..1'

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	[1,2,3,7,8,9,13,14,15,19,20,21]
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	[4,5,6,10,11,12,16,17,18,22,23,24]

Table 4: Wyckoff site: 2d, site symmetry: -6..1'

No.	position	mapping
1	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	[1,2,3,7,8,9,13,14,15,19,20,21]
2	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	[4,5,6,10,11,12,16,17,18,22,23,24]

Table 5: Wyckoff site: 4e, site symmetry: 3..1'

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,13,14,15]
2	$[0, 0, -z]$	[4,5,6,16,17,18]
3	$\left[0, 0, \frac{1}{2} - z\right]$	[7,8,9,19,20,21]
4	$\left[0, 0, z + \frac{1}{2}\right]$	[10,11,12,22,23,24]

Table 6: Wyckoff site: 4f, site symmetry: 3..1'

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,2,3,13,14,15]
2	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[4,5,6,16,17,18]
3	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	[7,8,9,19,20,21]
4	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	[10,11,12,22,23,24]

Table 7: Wyckoff site: 6g, site symmetry: .2.1'

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,4,13,16]
2	$[0,\ x,\ 0]$	[2,5,14,17]
3	$[-x,\ -x,\ 0]$	[3,6,15,18]
4	$\left[-x,\ -x,\ \frac{1}{2}\right]$	[7,10,19,22]
5	$\left[x,\ 0,\ \frac{1}{2}\right]$	[8,11,20,23]
6	$\left[0,\ x,\ \frac{1}{2}\right]$	[9,12,21,24]

Table 8: Wyckoff site: 6h, site symmetry: m..1'

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	[1,8,13,20]
2	$\left[-y,\ x-y,\ \frac{1}{4}\right]$	[2,9,14,21]
3	$\left[-x+y,\ -x,\ \frac{1}{4}\right]$	[3,7,15,19]
4	$\left[x-y,\ -y,\ \frac{3}{4}\right]$	[4,11,16,23]
5	$\left[y,\ x,\ \frac{3}{4}\right]$	[5,12,17,24]
6	$\left[-x,\ -x+y,\ \frac{3}{4}\right]$	[6,10,18,22]

Table 9: Wyckoff site: 12i, site symmetry: 11'

No.	position	mapping
1	$[x,\ y,\ z]$	[1,13]
2	$[-y,\ x-y,\ z]$	[2,14]
3	$[-x+y,\ -x,\ z]$	[3,15]
4	$[x-y,\ -y,\ -z]$	[4,16]
5	$[y,\ x,\ -z]$	[5,17]
6	$[-x,\ -x+y,\ -z]$	[6,18]
7	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	[7,19]
8	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[8,20]
9	$\left[-y,\ x-y,\ \frac{1}{2}-z\right]$	[9,21]
10	$\left[-x,\ -x+y,\ z+\frac{1}{2}\right]$	[10,22]
11	$\left[x-y,\ -y,\ z+\frac{1}{2}\right]$	[11,23]
12	$\left[y,\ x,\ z+\frac{1}{2}\right]$	[12,24]